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BIT 2223 Mobile Computing

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 7

Mobile IP

Objectives

By the end of this topic you should be able to;

- *Understand the Protocols for MOBILE networks*
- *Understand the TCP/IP protocol*
- *Understand the mobile IP protocol*
- *Understand the motivation of mobile IP design.*
- *Understand security considerations of mobile IP design*





7.1. Mobile Communication and Mobile Technology

Mobile phones are different size portable devices capable of communicating over longer distances through high tech processes involved know to be mobile computing and communications.

Today's mobiles networks supports features likes SMS, GPRS, MMS, emailing facility on mobile, Bluetooth, WAP and many more depending upon how reputed and bigger mobile network company is. Most of the networks world wide provides these features as they have become the standard features in mobile communication between their customers and of course one cannot neglect how sophisticated mobiles phones are available now, these mobile phones carries many features which some times are not supported by mobile networks. Mobile phones of today's age are now equal to portable PCs. These mobile phones connect to



BIT 2223 Mobile Computing

their cellular networks and these cellular networks are connected to Public switching telephone network (PSTN).

7.2. Mobile Network Technology

Mobile Technology has groomed a lot in past few years, major reasons for rapid advancements in mobile network technology is requirements for being mobile or connectivity on move. Latest mobile handsets offers features which one had never thought off, ultimately it forces mobile network companies to bring these features in practice use to take commercial advantages.

Cellular companies use AMPS, D-AMPS, CMMA2000, UMTS, GSM, EVDO etc. AMPS however pretty much vanished from the scene, AMPS network system was based on analog communication technology, latest features were not supported by



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AMPS therefore all cellular networks world wide have adopted above mentioned digital communication methodologies to meet the need of consumers. GSM remains the highly used mobile communication methodology worldwide. Cellular networks and mobile phones vary from geographical locations and providers to providers, but still standard communication methods are more or less same every where. Basic communication takes place using electromagnetic microwaves with cellular base stations. Cellular networks have huge antennas normally located in the middle of certain area to provide optimum signal broadcasting. These antennas are known as Base Transceiver Station (BTS). Mobile handsets have low powered transceivers which transmit voice data to the closed BTS which can usually be with in 5 t 8 miles radius.





Transceivers: Transceivers are devices which have capability of transmitting and receive data at the same time. Mobile handset register it self at mobile network switch or exchange as soon as it starts transmitting, usually it means when handset is turn on, it tries to registered to the network of inserted SIM card. Mobile networks uniquely identify each and every registered mobile handset on the given BTS and sends notifications to those mobile handsets which have incoming calls. Handsets will always be registered to the most nearest BTS for communication, incase if user of the network is on move, mobile phones in this case will handoff to many closest BTS during the call and keep the connection alive with nearest base stations. In case of unavailability of BTS, handset will be disconnected from the network resulting in dropping of on going call.



BIT 2223 Mobile Computing

BTS also operates on less power radio frequency transmission; they make their presence feel by broadcasting signals for receivers of mobile handsets. Communication between BTS and mobile sets takes place in stream of digital data, which is digitized audio. Each mobile network operates on unique radio frequency.

7.3. Wireless Transmission Protocols

There are several transmission protocols in wireless manner to achieve different application oriented tasks. Below, some of these applications are given.



7.3.1. Wireless Local Loop (WLL) and LMDS

Microwave wireless links can be used to create a wireless local loop. The local loop can be thought of as the "last mile" of the telecommunication network that resides between the central office (CO) and the individual homes and business in close proximity to the CO. An advantage of WLL technology is that once the wireless equipment is paid for, there are no additional costs for transport between the CO and the customer premises equipment. Many new services have been proposed and this includes the concept of Local Multipoint Distribution Service (LMDS), which provides broadband telecommunication access in the local exchange.

7.3.2. Bluetooth

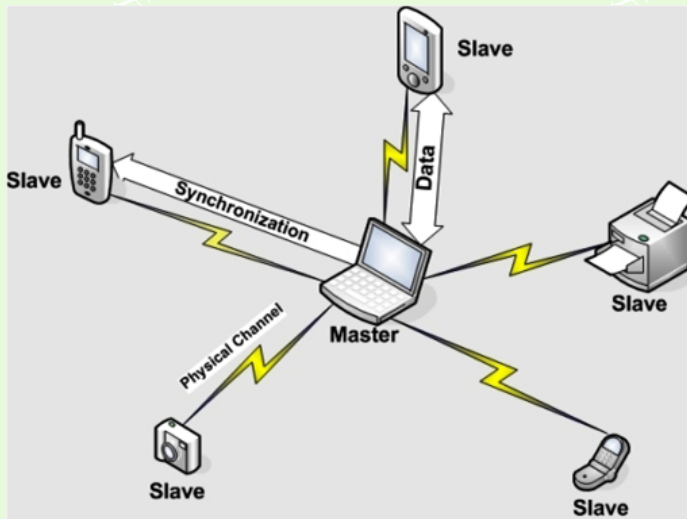


Fig: *Data transmission with Bluetooth.*

- Facilitates ad-hoc data transmission over short distances



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from fixed and mobile devices as shown in the above figure.

- Uses a radio technology called frequency hopping spread spectrum. It chops up the data being sent and transmits chunks of it on up to 79 different frequencies. In its basic mode, the modulation is Gaussian frequency shift keying (GFSK). It can achieve a gross data rate of 1 Mb/s
- Primarily designed for low power consumption, with a short range (powerclass- dependent: 1 meter, 10 meters, 100 meters) based on low-cost transceiver microchips in each device

7.3.3. Wireless Local Area Networks (W-LAN)

- IEEE 802.11 WLAN uses ISM band (5.275-5.825GHz)



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- Uses 11Mbps DS-SS spreading and 2Mbps user data rates (will fallback to 1Mbps in noisy conditions)
- IEEE 802.11a standard provides upto 54Mbps throughput in the 5GHz band. The DS-SS IEEE 802.11b has been called Wi-Fi. Wi-Fi networks have limited range. A typical Wi-Fi home router using 802.11b or 802.11g with a stock antenna might have a range of 32 m (120 ft) indoors and 95 m (300 ft) outdoors. Range also varies with frequency band.
- IEEE 802.11g uses Complementary Code Keying Orthogonal Frequency Division Multiplexing (CCK-OFDM) standards in both 2.4GHz and 5GHz bands.



7.3.4. WiMax

- Provides upto 70 Mb/sec symmetric broadband speed without the need for cables. The technology is based on the IEEE 802.16 standard (also called WirelessMAN)
- WiMAX can provide broadband wireless access (BWA) up to 30 miles (50 km) for fixed stations, and 3 - 10 miles (5 - 15 km) for mobile stations. In contrast, the WiFi/802.11 wireless local area network standard is limited in most cases to only 100 - 300 feet (30 - 100m)
- The 802.16 specification applies across a wide range of the RF spectrum, and WiMAX could function on any frequency below 66 GHz (higher frequencies would decrease the range of a Base Station to a few hundred meters in an



urban environment).

7.3.5. Zigbee

- ZigBee is the specification for a suite of high level communication protocols using small, low-power digital radios based on the IEEE 802.15.4-2006 standard for wireless personal area networks (WPANs), such as wireless headphones connecting with cell phones via short-range radio.
- This technology is intended to be simpler and cheaper. ZigBee is targeted at radio-frequency (RF) applications that require a low data rate, long battery life, and secure networking.
- ZigBee operates in the industrial, scientific and medical (ISM) radio bands; 868 MHz in Europe, 915 MHz in coun-



tries such as USA and Australia, and 2.4 GHz in most worldwide.

7.3.6. Wibree

- Wibree is a digital radio technology (intended to become an open standard of wireless communications) designed for ultra low power consumption (button cell batteries) within a short range (10 meters / 30 ft) based around low-cost transceiver microchips in each device.
- Wibree is known as Blue-tooth with low energy technology.
- It operates in 2.4 GHz ISM band with physical layer bit rate of 1 Mbps.



7.4. Mobile IP

Mobile IP is a communications protocol developed by the Internet Engineering Task Force. The design of this IETF protocol is configured to make it possible for consumers who use mobile communication devices to be able to move from one network to another, without having to make a change in their IP address. The function of Mobile IP allows the forwarding of Internet traffic to the mobile device even when the device is not connected to the home network. Authentication, authorization, and accounting represent the "big three" in terms of IP based network management and policy administration.

Authentication provides a vehicle to identify a client that requires access to some system and logically precedes authorization. The mechanism for authentication is typically undertaken



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through the exchange of logical keys or certificates between the client and the server.

Authorization follows authentication and entails the process of determining whether the client is allowed to perform and/or request certain tasks or operations. Authorization is therefore at the heart of policy administration.

Accounting is the process of measuring resource consumption, allowing monitoring and reporting of events and usage for various purposes including billing, analysis, and ongoing policy management.

The creation of Mobile IP meant a significant change in the way information terminates across the Internet. The routing of information depends on relaying data to IP addresses that are considered fixed in location. With the Mobile IP, instead of



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basing the termination on a fixed location associated with an IP address, the protocol calls for delivery of data to an IP address that may or may not be currently connected through the mother or home network.

Mobile IP is the underlying technology for support of various mobile data and wireless networking applications i.e. GPRS. Internet Protocol (IP) identifies each node uniquely, using an IP address that designates its physical attachment to the internet. It is a standard that allows users to move from one network to another without losing connectivity.

Mobile Computing is becoming increasingly important due to the rise in the number of portable computers and the desire to have continuous network connectivity to the Internet irrespective of the physical location of the node. The Internet infras-



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structure is built on top of a collection of protocols, called the TCP/IP protocol suite. Transmission Control Protocol (TCP) and Internet Protocol (IP) are the core protocols in this suite. IP requires the location of any host connected to the Internet to be uniquely identified by an assigned IP address. This raises one of the most important issues in mobility, because when a host moves to another physical location, it has to change its IP address. However, the higher level protocols require IP address of a host to be fixed for identifying connections.

Mobile devices have IP addresses that are associated with one network and moving to another network means changing IP address. Using the mobile IP system will allow users to achieve this and at the same time make the underlying process transparent for a user. IP executes in two (2) major steps: Packet routing



and packet forwarding.

1. **Packet routing:** Packet routing involves use of protocols like BGP, RTP and OSPF to decide the route that each packet has to travel.
2. **Packet forwarding:** It involves use of protocols like ARP, proxy ARP etc. to deliver the packet to the end node once it has arrived at the destination network.

In response to increasing popularity of palm-top and other mobile computers, Mobile IP was developed to maintain internet connectivity while moving from one Internet attachment point to another. Can work in either wired or wireless but more significant in wireless.



Mobile vs. nomadic

- Mobile: user is connected to one or more applications across the internet. User's point of attachment changes dynamically and all connections are automatically maintained despite the change.
- Nomadic: user's Internet connection is terminated each time the user moves and a new connection is initiated when the user dials back in software in the point of attachment is used (typically ISP) is used to obtain a new temporarily assigned IP address.

The Mobile Internet Protocol (Mobile IP) is an extension to the Internet Protocol proposed by the Internet Engineering Task Force (IETF) that addresses this issue. It enables mobile computers to stay connected to the Internet regardless of their lo-



cation and without changing their IP address. More precisely, Mobile IP is a standard protocol that builds on the Internet Protocol by making mobility transparent to applications and higher level protocols like TCP.

7.4.1. The TCP/IP Protocol Suite

TCP/IP protocol suite, the cornerstone of Internet networking, is a four-layer system. Each layer is responsible for a specific task. The four layers, from top to bottom, are application layer, transport layer, network layer, and link layer. The application layer handles the details of the particular application (e.g., FTP, TELNET, HTTP etc.). The transport layer provides a flow of data between two Internet nodes. There are two widely used transport layer protocols on the Internet: TCP (Transmission



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Control Protocol) and UDP (User Datagram Protocol).

TCP provides a reliable flow of data between two nodes by maintaining a connection-oriented environment. On the other hand, UDP provides an unreliable and connectionless datagram service. The network layer handles the movement of packets around the network by implementing efficient routing algorithms. IP (Internet Protocol), the default network layer protocol on the Internet, is described in detail in the next section. The link layer provides interfaces to the network hardware devices in the form of device drivers. Examples include IEEE 802.2 (LANs), X.25, packet radio etc. The physical layer, which is often tightly-coupled with the data link, is responsible for transmitting raw bits across the network through network interface cards and cables.





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Each layer provides some services that the upper layers use. Thus, support for mobility is likely to affect all the layers. The network layer that routes data to a destination host based on its location, needs to be modified so that it can handle routing when the physical location of the host changes. Similarly, at the transport layer, it is necessary to provide a better end-to-end delivery service, especially in the case of dropped packets; packets may be lost during mobility and need to be delivered immediately to the new location. Finally, the application layer requires additional support in terms of automatic configuration, service discovery, and link awareness. As an example of an application layer change, if an FTP session is in progress during mobility, the FTP application needs to configure itself being aware of the location changes. Mobile IP extends IP to support mobile com-



puting. The next lesson gives an overview of IP, as a preamble to Mobile IP.

7.4.2. Basics of IP addressing

All computers that are connected to the Internet need to have a valid IP address. This address is usually assigned by an Internet Service Provider (ISP) which in turn has bought a block of addresses from the Internet Cooperation for Assigned Names and Numbers (ICANN). Most companies never interact with the ICANN directly. In order for a company to receive valid IP addresses they contact a local ISP. Even local ISP:s do not interact with ICANN but in turn they contact larger ISP:s and only they contact ICANN.

IP addresses are not assigned randomly, but are carefully



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chosen. When a host wants to connect to the Internet he is assigned an IP address that represents the network that he is located at. This network is in turn a part of a larger network, which is then again a part of an even larger network, and in this way all networks are arranged, in a hierarchical manner. The IP address is divided into several parts where each corresponds to a given network. The rightmost part in the address always represents the host and the part to the left of it then represents the network that the host is located at and so on. In this way the leftmost part always represents the largest network. This hierarchy can be compared to a resident's home address.





7.4.3. Mobile IP Design

Each mobile node is assigned two IP addresses and transparently maintains the binding between the two addresses. One of the IP addresses is the permanent home address and the other is a temporary care-of address. The main goals of Mobile IP are to make mobility transparent to the higher level protocols and to make minimum changes to the existing Internet infrastructure. The two (2) addresses are explained below:

1. **Permanent Home Address:** This address is assigned at the home network and is used to identify communication endpoints.
2. **Care-of-address:** (Temporary). This address represents the current location of the host.



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The basic Mobile IP protocol has four distinct stages. These are:

1. **Agent Discovery:** Agent Discovery consists of the following steps:
 - a) Mobility agents advertise their presence by periodically broadcasting Agent Advertisement messages. An Agent Advertisement message lists one or more care-of addresses and a flag indicating whether it is a home agent or a foreign agent.
 - b) The mobile node receiving the Agent Advertisement message observes whether the message is from its own home agent and determines whether it is on the home network or a foreign network.
 - c) If a mobile node does not wish to wait for the periodic



advertisement, it can send out Agent Solicitation messages that will be responded by a mobility agent.

2. **Registration:** Registration consists of the following steps:

- a) If a mobile node discovers that it is on the home network, it operates without any mobility services.
- b) If the mobile node is on a new network, it registers with the foreign agent by sending a Registration Request message which includes the permanent IP address of the mobile host and the IP address of its home agent.
- c) The foreign agent in turn performs the registration process on behalf of the mobile host by sending a Registration Request containing the permanent IP address of the mobile node and the IP address of the foreign agent to the home agent.



- d) When the home agent receives the Registration Request, it updates the mobility binding by associating the care-of address of the mobile node with its home address.
 - e) The home agent then sends an acknowledgment to the foreign agent.
 - f) The foreign agent in turn updates its visitor list by inserting the entry for the mobile node and relays the reply to the mobile node.
3. **In Service:** This stage can be subdivided into the following steps:
- a) When a correspondent node wants to communicate with the mobile node, it sends an IP packet addressed to the permanent IP address of the mobile node.
 - b) The home agent intercepts this packet and consults the



mobility binding table to find out if the mobile node is currently visiting any other network.

c) The home agent finds out the mobile node's care-of address and constructs a new IP header that contains the mobile node's care-of address as the destination IP address. The original IP packet is put into the payload of this IP packet. It then sends the packet. This process of encapsulating one IP packet into the payload of another is known as IP-within-IP encapsulation, or tunneling.

d) When the encapsulated packet reaches the mobile node's current network, the foreign agent decapsulates the packet and finds out the mobile node's home address. It then consults the visitor list to see if it has an entry for that mobile node.



e) If there is an entry for the mobile node on the visitor list, the foreign agent retrieves the corresponding media address and relays it to the mobile node.

f) When the mobile node wants to send a message to a correspondent node, it forwards the packet to the foreign agent, which in turn relays the packet to the correspondent node using normal IP routing.

g) The foreign agent continues serving the mobile node until the granted lifetime expires. If the mobile node wants to continue the service, it has to reissue the Registration Request.

4. **Deregistration:** If a mobile node wants to drop its care-of address, it has to deregister with its home agent. It achieves this by sending a Registration Request with the



lifetime set to zero. There is no need for deregistering with the foreign agent as registration automatically expires when lifetime becomes zero. However if the mobile node visits a new network, the old foreign network does not know the new care-of address of the mobile node. Thus datagrams already forwarded by the home agent to the old foreign agent of the mobile node are lost.

7.4.4. Mobility binding - Home Agent (HA)

Mobile IP supports mobility by transparently binding the home address of the mobile node with its care-of address. This mobility binding is maintained by some specialized routers known as mobility agents. Mobility agents are of two types - home agents and foreign agents. The home agent, a designated router in the



Table 1: HA - Mobility binding table.

Home Address	Care-of Address	Lifetime (in sec)
131.193.171.4	128.172.23.78	200
131.193.171.2	119.123.56.78	150

Table 1: HA - Mobility binding table.



7.4.5. Mobility binding - Foreign Agent (FA)

Foreign agents are specialized routers on the foreign network where the mobile node is currently visiting. The foreign agent maintains a visitor list which contains information about the mobile nodes currently visiting that network. Each entry in the visitor list is identified by the tuple: < permanent home address, home agent address, media address of the mobile node, association lifetime>.

Home Address	Home Agent Address	Media Address	Lifetime (in s)
131.193.44.14	131.193.44.7	00-60-08-95-66-E1	150
131.193.33.19	131.193.33.1	00-60-08-68-A2-56	200



Table 2: FA Mobility binding table

7.4.6. Operation of Mobile IP

A router makes use of the IP address in an IP datagram to perform routing. Network portion of IP address is used by routers to move datagram from one computer to the network which the target computer is attached. The final router uses the host portion.

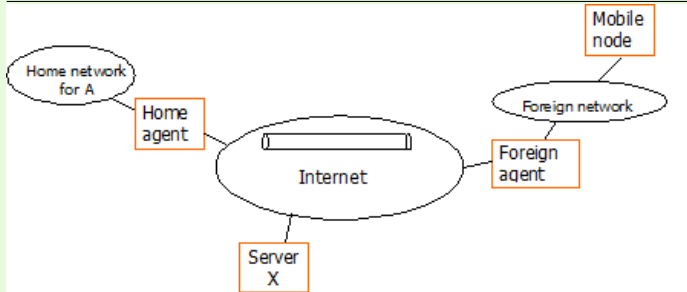
TCP establishes connection between communicating host mobile node, the IP address may change while one or more connection is active.

- Mobile node is assigned to a particular network known as home network.
- IP address on home network is static and is known as home



address

- Mobile node can move to another network known as foreign network
- Mobile node registers with network node on foreign network called foreign agent
- Mobile node then communicates with the similar agent on the users home network known as home agent, giving home agent care-of address that identifies the foreign agents location.



Exchange of IP datagram's between a mobile e.g. A and another host, e.g. server X

1. Server X transmits an IP datagram with A's home address on its IP header. Routed to A's home network
2. At home datagram intercepted by home agent. Home agent encapsulates datagram in another with A's care-of address in header and retransmits the datagram (The



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use of outer IP datagram with a different destination IP address is called tunneling). Datagram routed to foreign agent.

3. Foreign agent strips off the outer IP header, encapsulates the original IP datagram in a network-level PDU for delivery to A.
4. When A sends IP traffic to X, it uses X's IP address, sends to router on the foreign network for routing to X.

To support the mobile operations, mobile IP includes three capabilities.

7.4.7. Mobile IP capabilities

1. Discovery



- Mobile node uses discovery procedure to identify prospective foreign agents (or home agent).
- Mobile node is responsible for ongoing discovery process
- Must determine if it is attached to its home network or a foreign network
- Transition from home network to foreign network can occur at any time without notification to the network layer
- Mobile node listens for agent advertisement messages
- Compares network portion of the router's IP address with the network portion of home address

2. Registration

- mobile node uses an authenticated registration pro-



cedure to inform home agent of its care-of address.

- Once a MN is registered with Home agent, the home agent must be able to intercepts IP datagrams sent to mobile node's home address so that the datagram can be sent via tunneling.
- Home agent informs other nodes on home network that datagrams to mobile node should be delivered to home agent
- Datagrams forwarded to care-of address via tunneling
- Datagram encapsulated in outer IP datagram

3. Tunneling

- Home agent intercepts IP datagrams sent to mobile node's home address
- Home agent informs other nodes on home network that



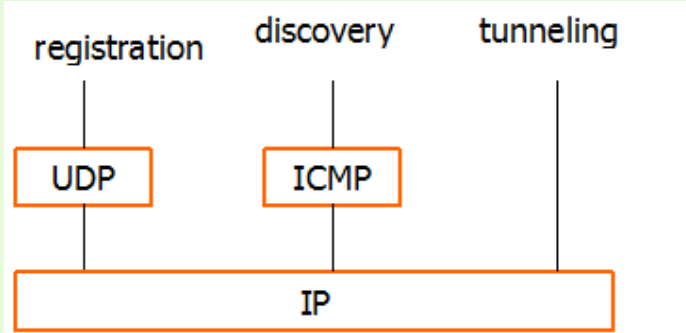
datagrams to mobile node should be delivered to home agent

- Datagrams are forwarded to care-of address via tunneling and is then encapsulated in outer IP datagram

7.4.8. Underlying protocols for mobile IP capabilities

Registration communicates between an application on the MN and application in HA and uses transport protocols. Since it is simple request response transaction, the overhead of connection oriented TCP is not required and therefore UDP is used. Discovery makes use of ICMP (internet control messaging protocol)

- Connectionless protocol suited for discovery.



- **Registration Process**

- Mobile node on foreign network, need to alert home agent
- Mobile node sends registration request to foreign agent requesting forwarding service
- Foreign agent relays request to home agent



- Home agent accepts or denies request and sends registration reply to foreign agent
- Foreign agent relays reply to mobile node

• Deregistration

If a mobile node wants to drop its care-of address, it has to deregister with its home agent. It achieves this by sending a Registration Request with the lifetime set to zero

7.5. Wireless Application Protocol (WAP)

It is an Open standard protocol providing mobile users of wireless terminals access to telephony and information services including Internet and web. Wireless terminals include wireless phones, pagers and personal digital assistants (PDAs). They



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are designed to work with all wireless network technologies such as GSM, CDMA, and TDMA. They are based on existing Internet standards such as IP, XML, HTML, and HTTP and includes security facilities.

- HTTP/HTML have not been designed for mobile devices and applications
- WAP empowers mobile users with wireless devices to easily access and interact with information and services.
- A “standard” created by wireless and Internet companies to enable Internet access from a cellular phone

DFN : Is a universal, Open standard providing mobile users of wireless phones and other wireless terminals such as pagers and PDA access to telephony and information services including Internet and web



- WAP is designed to work with all wireless network technologies such as GSM, CDMA, and TDMA
- Based on existing Internet standards such as IP, XML, HTML, and HTTP
- Includes security facilities
- WAP was established by WAP forum in 1997.

7.5.1. Limitations of mobile environment for data services

Mobile phones have significant device and network limitation for data services

1. Device limitations

- Processor



- Memory
- Battery power
- User interface(display is small)

2. Network limitations

- Low bandwidth
- High latency
- Unpredictable availability and stability

3. Wide variation of the above from device to device and network to network. WAP is designed to deal with these challenges.

7.5.2. WAP specifications

- A programming model (based on WWW programming model)



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- A markup language (adhering to xml) similar to HTML
- A specification of a small browser (micro browser) suitable for mobile phones and wireless terminals.
- A lightweight protocol stack
- A framework for wireless telephony applications (WTAs)
- Server-“Wap/Origin server”, similar to existing web servers
- Specifications define a set of protocols in application, session, transaction, security, and transport layers

7.5.3. WAP protocol stack

1. Application Layer

Wireless Application Environment (WAE). This layer is of most interest to content developers because it contains,



among other things, device specifications and the content development programming languages, WML and WMLScript.

2. Session Layer

Wireless Session Protocol (WSP). Unlike HTTP, WSP has been designed by the WAP Forum to provide fast connection suspension and reconnection.

3. Transaction Layer

Wireless Transaction Protocol (WTP). The WTP runs on top of a datagram service such as User Datagram Protocol (UDP) and is part of the standard suite of TCP/IP protocols used to provide a simplified protocol suitable for low bandwidth wireless stations.

4. Security Layer



Wireless Transport Layer Security (WTLS). WTLS incorporates security features that are based upon the established Transport Layer Security (TLS) protocol standard. It includes data integrity checks, privacy, service denial, and authentication services.

5. **Transport Layer**

Wireless Datagram Protocol (WDP). The WDP allows WAP to be bearer-independent by adapting the transport layer of the underlying bearer. The WDP presents a consistent data format to the higher layers of the WAP protocol stack, thereby offering the advantage of bearer independence to application developers.

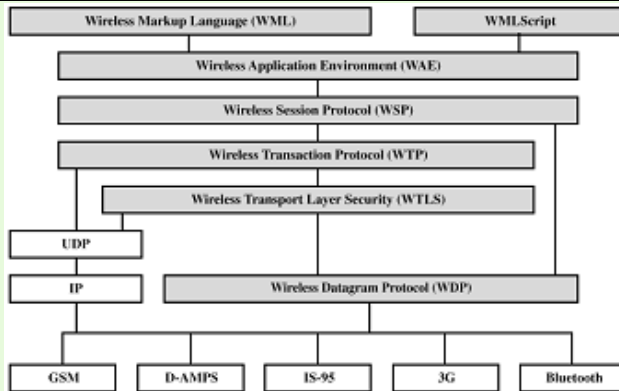


Fig. *WAP Protocol Stack*

WAP programming model is based on three elements:

- Original server,
- Gateway,



- Client

HTTP is used between gateway and original server to transfer content.

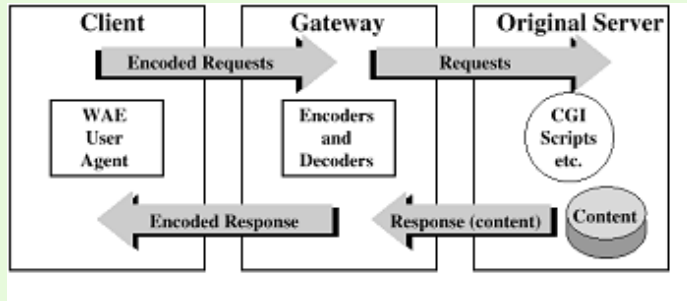


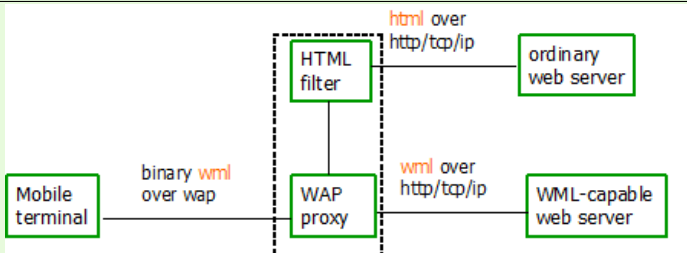
Fig: WAP Programming Model



- **Interactions**

A WAP Mobile user can browse content from web server. The web server provides contents in form of HTML coded pages that are transmitted using web protocol stack (HTTP/TCP/IP). The HTML content must go through the filter, which might be co-located in WAP proxy or in separate physical module.

Proxy converts HTML contents to WML. Proxy may also converts wml from WML compactable servers to more compact binary wml and delivers it to Mobile user.



7.6. Wireless Application Environment (WAE)

The part of the WAP protocol that application and service developers use most in their work. The WAE consists of the WML and WMLScript specs as well as the Wireless Telephony Application Interface (WTAI) that specifies how WAP applications can access mobile phone functionality (initiate a call, send an SMS). WAE specifies an application framework for wireless de-



vices

Gateway (act as proxy server): It is processor that provides services that offload limited capabilities of hand-held, mobile and wireless terminals, e.g.it provides provide DNS services, Convert between WAP and WWW stack. Encode information from web into more compacted form and on the other direction decode compacted info to standard web conventions. Gateway also Cache frequently requested info.

7.6.1. WAE elements:

- WAE User agents: Software that executes in the wireless device
- Content generators: Applications that produce standard content formats in response to requests from user agents



in the mobile terminal

- Standard content encoding: Defined to allow a WAE user agent to navigate Web content
- Wireless telephony applications (WTA): Collection of telephony-specific extensions for call and feature control mechanisms



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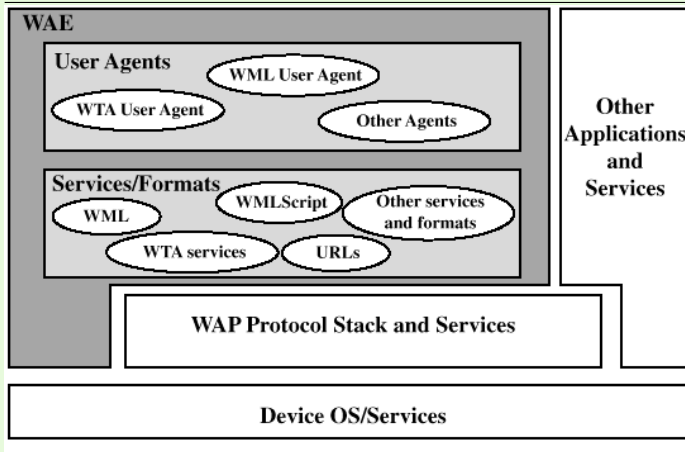


Fig. *WAE Client Components*

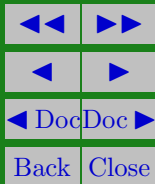


7.7. Wireless Markup Language (WML)

It was designed to describe contents and format of presenting data on devices that are limited to bandwidth, screen size and user input capabilities. When HTML is converted to wml graphics and animations are stripped away. Presents text based information.

7.7.1. Important features of wml

- Text and image support: formatting and layout commands are provided for text and limited image capabilities.
- Deck/card organizational metaphor: WML documents subdivided into cards, which specify one or more units of interaction, e.g. a menu, a screen of text, or a text entry field). Wml deck is similar to HTML page in that it is identified





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by URL address and is the unit of content transmission.

- Support for navigation among cards and decks: Includes provisions for event handling; used for navigation or executing scripts.

WML is tagged language, similar to HTML. WML definition of card begins with non visible portion which contains executable elements followed by the visible part.

```
<wml>
<card id='card1'>
<p>
Hello wap world
</p>
<p>.....</p>
</card></wml>
```



7.7.2. WMLScript

Scripting language for defining script-type programs in a user device with limited processing power and memory (similar to JavaScript)

- **WMLScript capabilities:**

- Check validity of user input before it's sent
- Access device facilities and peripherals
- Interact with user without introducing round trips to origin server

{

- WMLScript is the scripting language used in WML pages
- WMLScript is a light version of the JavaScript language



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- WML scripts are not embedded into WML pages
- WML pages contains references to script URLs
- WMLScript is compiled into byte code on the server before it is sent to the WAP browser
- WMLScript is a part of the WAP specification

}

- **WMLScript features:**

- JavaScript-based scripting language
- Procedural logic
- Event-based
- Compiled implementation
- Integrated into WAE



7.8. Wireless Session Protocol (WSP)

It is a Transaction-oriented protocol based on the concept of a request and a reply (browsing session). It provides applications with interface for two session services:

- Connection-oriented session service: operates above reliable transport protocol WTP
- Connectionless session service: operates above unreliable transport protocol WDP

7.8.1. Connection-mode WSP Services

It establishes reliable session from client to server and release (like TCP). It agrees on common level of protocol functionality using capability negotiation. It exchanges content between



client and server using compact encoding. It also suspends and resumes a session and pushes content from server to client in an un-synchronized manner.

7.8.2. WSP Transaction Types

1. Session establishment – client WSP user requests session with server WSP user
2. Session termination – client WSP user initiates termination
3. Session suspend and resume – initiated with suspend and resume requests
4. Transaction – exchange of data between a client and server
5. Non-confirmed data push – used to send unsolicited information



mation from server to client

6. Confirmed data push – server receives delivery confirmation from client

7.9. Wireless Transaction Protocol (WTP)

It is a lightweight protocol suitable for "thin" clients and over low-bandwidth wireless links. It is a transaction oriented rather than a connection oriented protocol

7.9.1. WTP features

- Has three classes of transaction service
 - Optional user-to-user reliability: WTP user triggers confirmation of each received message
 - Optional out-of-band data on acknowledgments



- PDU concatenation and delayed acknowledgment to reduce the number of messages sent
- Asynchronous transactions

7.10. Wireless Transport Layer Security (WTLS)

It provides security services between client and WAP gateway, based on TLS protocol.

7.10.1. WTLS Features

- **Data integrity:** Ensures that data sent between client and gateway are not modified, using message authentication
- **Privacy:** Ensures that the data cannot be read by a third party, using encryption

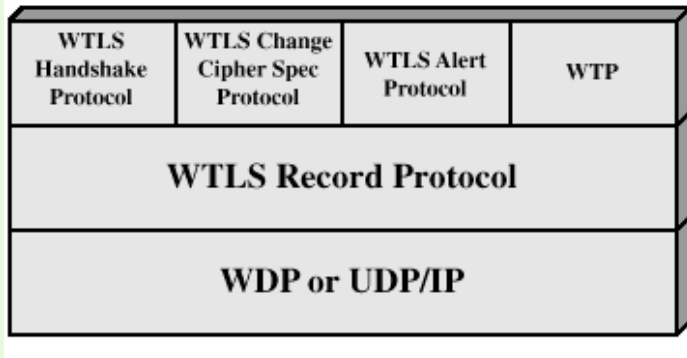


- **Authentication:** Establishes authentication of the two parties, using digital certificates
- **Denial-of-service protection:** Detects and rejects messages that are replayed or not successfully verified

7.10.2. WTLS Protocol Stack

WTLS consists of two layers of protocols

1. WTLS Record Protocol: provides basic security services to various higher-layer protocols
2. Higher-layer protocols:
 - The Handshake Protocol
 - The Change Cipher Spec Protocol
 - The Alert Protocol



7.10.3. Phases of the Handshake Protocol Exchange

1. First phase: Used to initiate a logical connection and establish security capabilities
2. Second phase: Used for server authentication and key exchange



3. Third phase: Used for client authentication and key exchange
4. Forth phase: Completes the setting up of a secure connection

7.11. Wireless Datagram Protocol (WDP)

- Used to adapt higher-layer WAP protocol to the communication mechanism used between mobile node and WAP gateway
- WDP hides details of the various bearer networks from the other layers of WAP
- Adaptation may include:
 - Partitioning data into segments of appropriate size



for the bearer

- Interfacing with the bearer network

7.12. Wireless Control Message Protocol (WCMP)

- Performs the same support function for WDP as ICMP does for IP
- Used in environments that don't provide IP bearer and don't lend themselves to the use of ICMP
- Used by wireless nodes and WAP gateways to report errors encountered in processing WDP datagrams
- Can also be used for informational and diagnostic purposes



7.13. VOIP Network

VoIP is a combination of two technologies Voice data and Internet Protocol to transmit voice on longer distances with lowest costs. VoIP technology determines how Internet telephony, making calls on internet, making calls from phone to PC, Pc to Phone and Phone to Phone works at present day. VoIP is a technology which is used to transmit voice traffic, (voice data) over IP network, in such kind of networks specially designed voice applications are used for transferring voice data, one can quote an example of such voice applications are voice chat, Internet phones which we used after buying minutes which are very inexpensive and teleconferencing.

VoIP technology is quite dynamic in nature and it has potential to provide solutions for every layer of internet protocol



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network. IP was first invented for data communication over computer networks and to connect all the computers together forming one big gigantic network called Internet. The potential of Internet protocol in regard to vast possible applications led to voice communication networking called VoIP or Voice over Internet Protocol. Due to the potential of Voice of Internet protocol, it has been one of the fastest growing industries in recent times when we talk about voice communication and telephony. There is lot of work going on using this technology and plenty of room is left to utilize to its full potential for obtaining efficiency and cost effect voice communication solutions.



7.13.1. Voice over IP and Mobile Communication

VoIP is also playing significant role in Mobile communication technologies and giving competitive edge in businesses between companies who have deployed VoIP services in their mobile network. You must have noticed in recent times about calling charges are way much more less than if you compare it with past, every thing else in getting expensive where as calls are getting cheaper and cheaper, it is all due to voice over internet protocol. With the help of VoIP networking; internationals calls are way much cheaper now and they cost almost as much as local calls, more and more free minutes are provided to get business benefits by different mobile companies to attract more consumer market.

Calling cards is another successful method for voice commu-



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nication which uses VoIP networks. One can call any number in any where in the world at very less price with good voice quality by purchasing calling cards. The VoIP based services are easily available for residential and corporate sectors; voice communication can be made by PC to PC, PC to phone and phone to phone.

There are many other practical features of VoIP technology.

1. Making calls from one end of the world to the other end of the world with very nominal cost price. Broadband however is required for VoIP network to work. Subscribers of VoIP phones can take IP phones along with them any where in the world for staying in touch by cost effective communication back home.
2. It supports software phones as well, these are also called



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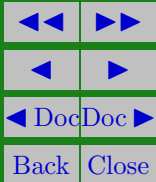
soft phones. Soft phones are basically software which interfaces exactly like phone on computer for communication. Using such soft phones one can make phone calls every where in the world, Soft phones are connected to VoIP network with help of broadband internet connections.

3. Regular telephony network like call waiting, call transfer, conference calling, caller Id, etc, there may be value added features like call filtering, call forwarding, sending calling directly to voice etc, such services might come with extra price by VoIP service providers.
4. Ability to check voice mail from remote location or sending email with attachments using internet.
5. VOIP costs a lot less to call thousands of miles from your location if you compare it from your landline numbers or

regular mobile phone.



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Revision Questions

EXERCISE 1.  Explain capabilities of Mobile IP

Example . List the WMLScript capabilities

Solution:

Check validity of user input before it's sent

Access device facilities and peripherals

Interact with user without introducing round trips to origin
server ☐

EXERCISE 2.  Explain WTLS features



Solutions to Exercises

Exercise 1.

Discovery – mobile node uses discovery procedure to identify prospective home and foreign agents.

Registration – mobile node uses an authenticated registration procedure to inform home agent of its care-of address.

Tunneling – used to forward IP datagrams from a home address to a care-of address

Exercise 1



Exercise 2.

Data integrity: Ensures that data sent between client and gateway is not modified and uses message authentication

Privacy: Ensures that the data cannot be read by a third party and it uses encryption

Authentication: Establishes authentication of the two parties and it uses digital certificates

Denial-of-service protection: Detects and rejects messages that are replayed or not successfully verified

Exercise 2